

MODULE 06 · DOMAIN 1 - DETECTION

Why the log is not there

Troubleshooting detection – the short list of reasons a scenario contains no evidence at all.

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Logging is a chain of four links. **Generate, permit, deliver, match.** Any one of them silently breaks it.

Six causes, and the tell for each

CAUSE	THE TELL IN THE SCENARIO
Never enabled	The scenario names precisely which logging is on
Service lacks permission to write	It worked, then a policy was tightened
Wrong Region or account	Multi-Region, or a global resource, or an org
The agent is not reporting	EC2 or on-premises, and only some hosts are missing
The filter never matched	The data exists but nothing fired
It was there and got removed	Retention, lifecycle, or a suppression rule

ON THE EXAM

Every one of these produces the same symptom: nothing. The tell is in the scenario's wording, not in the symptom.

What logs by default, and what does not

SOURCE	DEFAULT
CloudTrail management events	ON. Event history, 90 days
CloudTrail data and network activity events	OFF
VPC Flow Logs	OFF. And you choose ACCEPT, REJECT or ALL
Route 53 Resolver query logging	OFF
S3 server access logging	OFF
ELB, CloudFront and WAF logging	OFF
API Gateway execution and access logs	OFF, and needs an account-level role
Lambda to CloudWatch Logs	ON, if the execution role permits it

ON THE EXAM

Almost everything is off. That single fact resolves more troubleshooting questions than any other on this exam.

The awkward per-service switches

- ◆ **API Gateway:** the CloudWatch role ARN is set on the **account, per Region**, for every REST and WebSocket API in it.
- ◆ **AWS WAF:** the logging destination name **must start with aws-waf-logs-**.
- ◆ A WAF log group must match the web ACL's account and Region. For **CloudFront**, that means **us-east-1**.
- ◆ One destination per web ACL. WAF creates the resource policy or bucket policy itself.
- ◆ **VPC Flow Logs:** the traffic type filter. ACCEPT will never show you a rejection.

ON THE EXAM

Arbitrary facts, which is exactly why they get examined.

The delivery permission chain

- ◆ **To S3:** a bucket policy allowing the delivering service principal. Tighten it carelessly and delivery stops.
- ◆ **To CloudWatch Logs:** a resource policy on the log group, or a role the service assumes. VPC Flow Logs uses a role.
- ◆ **To an encrypted destination:** the KMS key policy must permit the service. This is the quiet one.
- ◆ Encrypt permission without decrypt permission means logs arrive and nobody can read them.
- ◆ Symptom to remember: **it worked, someone tightened a policy, now it does not.** Three policies can each cause it.

ON THE EXAM

When both reads and writes fail at once, suspect the key policy. When only reads fail, suspect the missing Decrypt.

Scope: Region, account and global resources

- ◆ **Almost everything is Regional** – Config recorder, GuardDuty, Security Hub, log groups, single-Region trails.
- ◆ **Global** resources log somewhere specific. CloudFront and its WAF log in **us-east-1**.
- ◆ CloudTrail replicates **global service events** into every Region you have enabled.
- ◆ In an organisation, ask **which account** receives it. An unreadable archive is the same as no log.
- ◆ New accounts need **auto-enable**. New Regions need the trail to be **multi-Region**.

ON THE EXAM

'We looked in eu-west-1' fully explains a missing CloudFront log.

Agents and execution roles

- ◆ **CloudWatch agent on EC2** needs an instance profile with logging permissions – usually **CloudWatchAgentServerPolicy**.
- ◆ Its config usually lives in **SSM Parameter Store**, so the instance must be **SSM managed**.
- ◆ Usual failures: agent never started, wrong log file path, unparseable timestamp.
- ◆ **Lambda**: the execution role needs **CreateLogGroup, CreateLogStream, PutLogEvents**. Without them it runs and logs nothing.
- ◆ Agent problems hit **some hosts**. Policy problems hit **all** of them.

ON THE EXAM

AWSLambdaBasicExecutionRole grants exactly those three actions. That is why it exists.

It matched nothing, or it was removed

- ◆ **Metric filters are not retroactive.** Created today, never matches yesterday.
- ◆ **Infrequent Access has no metric extraction**, so the alarm sits in INSUFFICIENT_DATA.
- ◆ A plain-text pattern will not match JSON. CloudTrail needs the `{ $.field = "value" }` form.
- ◆ **Retention** deleted it, or **lifecycle** moved it to a tier you did not query.
- ◆ **GuardDuty suppression rules** archive findings, removing them from attack sequence correlation.

ON THE EXAM

'The data exists' and 'the alert fired' are two different claims.

The order to work through it

01

Generated?

Is that log source actually switched on, with the right filter?

02

Permitted?

Bucket policy, log group policy, service role, KMS key policy.

03

Where?

Right Region, right account. Global resources log in us-east-1.

04

Reporting?

Agent running, SSM managed, instance profile correct.

05

Matched?

Filter pattern, log class, retention, suppression rules.

ON THE EXAM

Work top down. The first four are cheap to check and account for most failures. Only then start reading filter patterns.

A retailer runs REST APIs on API Gateway in eu-west-1 and us-east-1. After a fraud incident, the team enables execution logging at INFO level on the eu-west-1 payments stage and enables custom access logging to a new log group.

No log group named API-Gateway-Execution-Logs appears, and the access log group stays empty, although the API is serving traffic normally. The identical configuration on the us-east-1 stage began producing logs immediately. Both stages were configured by the same engineer within an hour.

What is the MOST likely cause?

- ☐ A The eu-west-1 log group was created after the stage, so the stage must be redeployed
- ☐ B Execution logging at INFO level only records errors, so nothing was written
- ☐ C The API Gateway service-linked role was not created in eu-west-1
- ☐ D The account-level CloudWatch log role ARN is not set for API Gateway in eu-west-1

◆ PAUSE AND DECIDE

What is the MOST likely cause?

- A

 The log group predates the stage; redeploy
Invented mechanics. Order of creation does not matter, and a redeployment would not create a permission that was never granted.
- B

 INFO only records errors
Backwards. ERROR records only failed requests; INFO records all of them. And it would not explain the empty access log group.
- C

 The service-linked role was not created
The right shape of answer with the wrong mechanism. API Gateway CloudWatch logging uses a role you create and nominate, not a service-linked role.
- D

 The account-level CloudWatch role ARN is unset in eu-west-1
Correct. That setting is per account per Region and governs every REST and WebSocket API in the Region. us-east-1 works because it was set there.

THE SIGNAL

'It works in one Region and not another, same config, same engineer' is the signature of a Region-scoped account setting.

A fintech deploys a new Lambda function that validates payment instructions. The team wrote a tightly scoped execution role by hand rather than using a managed policy, granting only the DynamoDB and KMS actions the function needs.

The function runs correctly in production for three weeks. During an investigation, the security team finds no CloudWatch log group for the function at all. There were no deployment errors, no failed invocations, and CloudWatch metrics for the function show invocations and durations normally.

Why are there no logs?

- ☐ A Lambda logging must be enabled explicitly in the function's configuration
- ☐ B The log group's retention policy deleted the streams before the investigation
- ☐ C The execution role lacks logs:CreateLogGroup, CreateLogStream and PutLogEvents
- ☐ D Lambda logs are only retained when the function is invoked inside a VPC

◆ PAUSE AND DECIDE

Why are there no logs?

- A

 Logging must be enabled in the function configuration
Lambda logs by default. There is no switch to turn on – the only requirement is that the execution role permits the three logs actions.
- B

 Retention deleted the streams
Retention deletes streams inside a log group. The scenario says there is no log group at all, which is a different failure entirely.
- C

 The execution role lacks the three logs actions
Correct. Lambda creates the log group on first write using the execution role's permissions. No permission, no log group, and no error – logging failures do not fail an invocation.
- D

 Logs need VPC invocation
Fabricated. VPC configuration affects network access from the function; it has nothing to do with CloudWatch Logs delivery.

THE SIGNAL

'No log group at all' is stronger than 'no logs'. A missing group means nothing ever had permission to create it.

A government contractor is investigating whether an EC2 instance attempted to reach blocked internal subnets. VPC Flow Logs were enabled on all VPCs eighteen months ago and deliver to CloudWatch Logs, where the log groups contain millions of records.

Queries for the instance's ENI return plenty of records, all of them showing accepted connections to expected destinations. The security team knows from network firewall logs that this instance made hundreds of connection attempts that were denied.

Why do the flow logs show no denied attempts?

- ☐ A The flow logs were created with a traffic type filter of ACCEPT
- ☐ B VPC Flow Logs record only traffic that leaves the VPC
- ☐ C The log group's retention period expired the REJECT records first
- ☐ D Denied traffic is recorded by Network Firewall, never by VPC Flow Logs

◆ PAUSE AND DECIDE

Why do the flow logs show no denied attempts?

- A

 The traffic type filter was set to ACCEPT
Correct. Flow logs capture ACCEPT, REJECT or ALL, chosen at creation. Set to ACCEPT, rejected traffic is never recorded and there is no gap to notice.
- B

 Flow logs only record traffic leaving the VPC
False. Flow logs capture traffic to and from network interfaces, including entirely internal VPC traffic.
- C

 Retention expired the REJECT records
Retention deletes by age, not by record type. It cannot remove rejections while leaving contemporaneous accepted records in place.
- D

 Only Network Firewall records denied traffic
Flow logs record rejections perfectly well when configured to. The presence of firewall logs is corroboration, not an explanation.

THE SIGNAL

'Plenty of records, all of them accepted' is not a gap. It is a filter, described precisely.

A media company streams application logs from 300 EC2 instances to CloudWatch Logs using the CloudWatch agent, with the configuration held in SSM Parameter Store and applied through a State Manager association.

After an Auto Scaling event replaces part of the fleet, 40 of the new instances send no logs. The other 260, including some launched the same day, report normally. All instances use the same AMI and the same launch template. The instances are reachable and the application is serving traffic.

What should the team investigate FIRST?

- ☐ A The CloudWatch Logs resource policy, which may be rejecting writes from the new instances
- ☐ B Whether those instances are SSM managed, so the agent configuration was applied
- ☐ C The log group retention period, which may have been shortened
- ☐ D Whether the log group was moved to the Infrequent Access class

◆ PAUSE AND DECIDE

What should the team investigate FIRST?

- A

 The CloudWatch Logs resource policy
A policy problem is account-wide. It would silence all 300 instances, not 40. The partial failure rules this out immediately.
- B

 Whether those instances are SSM managed
Correct. The agent config arrives via SSM, so an instance that never registered gets no config. Partial failure on identical instances is the agent-shaped tell.
- C

 The log group retention period
Retention affects the log group, so it would affect every instance writing to it. It also deletes old data rather than preventing new writes.
- D

 The Infrequent Access log class
A real trap from module two, and it is a property of the log group. It would break metric extraction for everyone, not ingestion for 40 hosts.

THE SIGNAL

40 of 300, identical AMI and launch template, is the whole question. Partial failure means the host, not the destination.

Five things to carry forward

- ◆ **Almost everything is off by default.** Check whether the source was ever generating before anything else.
- ◆ **Match the scope of the cause to the scope of the symptom.** Some hosts means the host. All hosts means the destination.
- ◆ **API Gateway's CloudWatch role is per account, per Region.** Works in one Region, not another, same config.
- ◆ **Lambda logs nothing if the execution role omits the three logs actions,** and nothing fails.
- ◆ **A flow log full of accepted traffic is a filter, not a gap.** Healthy-looking logs can be half missing.

ON THE EXAM

That closes Domain 1. Next: Domain 2, incident response – designing the plan before you need it.